

CSE4502 Operating Systems Lab 2A

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1 Part A: File Permission Checker (Conditionals and String Operations)

1.1 Problem Statement

Develop a Bash script named `check_permissions.sh` that checks whether a given file is readable, writable, and executable by the current user.

1.2 Logic

1. Take a filename as a command-line argument.
2. Verify if the file exists using `[ -e $filename ]`.
3. If it exists, check each permission using `-r`, `-w` or `-x`
4. Display the permission status clearly.

1.3 Hints

- Use `echo` statements to print messages for each permission type.

1.4 Example Output

```
$ ./check_permissions.sh script.sh
File: script.sh
Readable: Yes
Writable: Yes
Executable: No
```

2 Part B: Concurrent Multi-Account Transfer System

2.1 System Description

Three shared bank accounts are initialized with the following balances:

$$A = 1000, \quad B = 1000, \quad C = 1000$$

Three threads execute concurrently, each performing a fixed number of money transfer transactions between the accounts.

2.2 Workload Specification

The workload consists of the following concurrent threads:

- **Thread T1:** Transfers $Amount_1$ units from account A to account B , repeated $Repeat_1$ times.
- **Thread T2:** Transfers $Amount_2$ units from account B to account C , repeated $Repeat_2$ times.
- **Thread T3:** Transfers $Amount_3$ units from account C to account A , repeated $Repeat_3$ times.

Each repetition is considered a distinct transaction. Each $Amount_i$ and $Repeat_i$ are integers. And the relationship between $Amount_i$ and $Repeat_i$ is given as $Amount_i * Repeat_i = 1000$ and $Amount_i \mid 1000$ that is 1000 is divisible by $Amount_i$.

2.3 Correctness Constraints

The system must satisfy the following constraints:

- Each transaction must be **atomic** (all-or-nothing).
- No account balance may ever become negative.
- Concurrent execution must not violate consistency.

2.4 Expected Input, Output and Execution Behavior

2.5 Input

Input should be of 3 integers, as such:

Amount1 Amount2 Amount3

And each of the input integers satisfy the constraint $Amount_i * Repeat_i = 1000$.

2.5.1 Transaction Logging

For every transaction, the system must generate logs indicating the completion of the transfer.

[Tk] Transferred <amount> from <source> to <destination>

2.5.2 Intermediate Account States

After each thread completes all of its assigned transactions, the system must print the current account balances:

After T1 completes: $A = x1, B = y1, C = z1$

After T2 completes: $A = x2, B = y2, C = z2$

After T3 completes: $A = x3, B = y3, C = z3$

After each thread completes the expression $A + B + C = 3000$ must hold

2.5.3 Final System State

After all threads have completed execution, the final account balances must be:

Final Balances:

$A = 1000$

$B = 1000$

$C = 1000$